

ChargEV: Find. Charge. Go Green.

The Smart EV Charging Station Platform
Project Proposal for UCS503P
Submitted to: Nisha Thakur
Thapar Institute of Engineering and Technology

Pehal, CSED*

Briti Arora, CSED†

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1 Higher-order goal

...secondary framing

This project aims to accelerate the adoption of electric vehicles (EVs) by removing a major piece of charging infrastructure friction: the uncertainty around where to charge, whether a station is available, and whether a slot can be guaranteed. While the broader goal is a smarter, greener transportation ecosystem, the immediate engineering objective is to translate *predictable, low-anxiety EV charging* into a measurable, deployable software solution.

2 Time-to-value

...primary heuristic

- We will deliver meaningful increments quickly by prototyping the core discover-book-charge workflow and validating it with users within a short iteration window.
- We will prioritize fast feedback over perfection by using weekly iterations (and targeting CI-backed automation early) to reduce release friction.
- Success is defined by what can be delivered and measured in the first iteration – a working station locator and booking flow – then improved based on user feedback.

3 Problem Statement

EV drivers routinely face friction that turns charging into a source of anxiety rather than a routine task. Common pain points include:

- **Discovery gap:** it is hard to find nearby charging stations from a single, reliable source.
- **Unclear availability:** station status (free, occupied, out of service) is rarely known before arriving.
- **Wasted time:** drivers wait at busy stations with no way to reserve a slot in advance.
- **Generic search results:** existing tools do not filter by connector type, distance, availability, or price together.

*Roll No: 1024170103, Email: ppehal_be24thapar.edu

†Roll No: 1024170431, Email: barora2_be24thapar.edu

- **Difficult trip planning:** long-distance EV travel requires manually chaining together charging stops.
- **Low engagement:** there is little incentive for drivers to keep returning to a single platform.

Why this matters now (contextual relevance): as EV ownership grows, even small improvements in charging predictability translate into measurable reductions in range anxiety and wasted driver time.

4 Proposed Solution

... Software Proposition

4.1 Overview

Build a **web-based** EV charging platform, *ChargEV*, with the following components:

- **Centralized station locator:** a single source of truth for nearby EV charging stations.
- **Real-time availability dashboard:** live status per station/connector.
- **Online slot booking & reservation system:** drivers reserve a charging slot in advance.
- **Smart search & filters:** filter by connector type, distance, availability, and price.
- **Trip planning with charging stops:** route-aware suggestions for long-distance travel.
- **Rewards, badges & eco-impact tracking:** gamified engagement layer (“EcoCoins”).

4.2 Core workflow

The user journey is organized into three phases:

1. **Discover:** access the cross-device web platform, search/filter nearby stations, and evaluate details such as live availability, pricing, connector compatibility, and reviews.
2. **Act:** book a time slot and confirm instantly, then get seamless routing/navigation to the station for a guaranteed charge.
3. **Engage:** accumulate EcoCoins and track personal savings and eco-impact, incentivizing return usage.

4.3 Operational constraints for an educational setting

- Keep the solution usable on low-to-mid range devices via responsive web design.
- Avoid dependence on advanced research algorithms; focus on workflow, real-time data, and system correctness.
- Design for deployability using common web hosting, managed databases, and existing location-intelligence APIs.

5 Solution Approach

... Engineering focus

This is an engineering course project, so we focus on a practical, scalable web architecture rather than novel algorithm research.

5.1 System architecture

... *web-based solution*

A typical 3-tier web architecture:

- **Frontend (the interface):** built for speed and cross-device responsiveness using HTML5, CSS3, Bootstrap 5, and JavaScript.
- **Backend (the engine):** high-performance, event-driven request routing using Node.js and Express.js.
- **Data & logistics (the brains):** cloud-native storage via MongoDB Atlas, combined with location-intelligence APIs – Google Maps API, Google Places API, and Google Directions API – for search, routing, and trip planning.

5.2 CI/CD and fast delivery enabler

CI/CD will be used to:

- Automatically build and run tests on every change.
- Produce repeatable deployment artifacts to staging.
- Enable safe and frequent releases of incremental features (e.g., filters, then trip planning, then rewards).

6 Evaluation Criterion

... *measurable and attributable*

We define success using measurable metrics tied to the core discover-book-charge workflow.

6.1 Primary evaluation metric

Time-to-charge (TTC): time between a user opening the app to find a station and successfully starting a charge at a booked slot.

- **Target:** median TTC reduced compared to a no-booking baseline (walk-in search).
- **Attribution:** measured from application timestamps (search start vs. booking confirmation vs. charge start).

6.2 Secondary evaluation metrics

- **Booking conversion rate:** percentage of searches that convert into a confirmed slot booking.
- **Availability data accuracy:** agreement between displayed status and actual station status at arrival.
- **Station utilization:** change in booked-slot usage for participating stations.
- **User engagement:** repeat-usage rate and EcoCoins accumulation per active user.

6.3 Pilot validation plan

... *high-level*

- Recruit 10–20 pilot EV users and coordinate with 2–3 charging station operators (simulated if real operators are unavailable).
- Compare metrics before/after within the iteration window by logging baseline estimates via short interviews or existing walk-in behaviour.

7 Scalability

... with foresight

The proposition should scale in both theoretical and practical senses.

1. **Scalable (theoretically):** the system should support growth in the number of stations, bookings, and concurrent users by:
 - decoupling components (frontend/backend),
 - enabling pagination and indexing for common location queries,
 - using a stateless, event-driven API design.
2. **Operationally deployable (practically):** the system should run on common web hosting:
 - cloud-native database (MongoDB Atlas),
 - containerized or platform-hosted deployment,
 - CI/CD pipeline for staging/production.

8 Engine availability heuristic

A mature set of “engines” (tooling/services) is available to implement this as a web-based project:

- Web framework for REST APIs (Node.js, Express.js).
- Managed cloud document database (MongoDB Atlas).
- Location-intelligence APIs (Google Maps, Places, Directions) for search, geocoding, and routing.
- Authentication approach (token-based or session-based).
- Test framework for unit/integration tests.
- CI runner and staging deployment target.

We will use these mature components to focus on workflow design, real-time correctness, and measurement rather than inventing new infrastructure.

9 Project scope and deliverables

... iteration-friendly

9.1 Initial deliverable

... first iteration

- Centralized EV charging station locator (search + map view)
- Real-time availability and status dashboard
- Online slot booking & reservation flow
- Smart search & filters (connector type, distance, availability, price)
- CI: build + backend unit tests + linting
- CD to staging with a single command or pipeline job

9.2 Subsequent deliverables

- Trip planning with charging stops for long-distance routes
- Rewards, badges & eco-friendly impact tracking (EcoCoins)
- Admin dashboard for station operators (utilization, schedules)
- Improved search/filtering and indexed queries for scale readiness

10 Risks and mitigations

- **Stale or incorrect availability data:** rely on periodic sync with station status and clearly timestamp last-updated status; flag stale data in the UI.
- **Third-party API costs/limits:** cache Google Maps/Places/Directions responses where possible and monitor quota usage.
- **Low initial station coverage:** start with a curated set of stations/operators for the pilot and expand incrementally.
- **User adoption:** keep the booking flow to a minimal number of steps and use the EcoCoins/rewards loop to encourage repeat use.
- **Location data privacy:** minimize stored location data; only persist what is needed for search and trip planning.

11 Summary

This project proposes ChargeEV, a deployable web platform that eliminates range anxiety by making EV charging simple, convenient, and predictable through a centralized locator, real-time availability, and smart online booking. It meets the course criteria by emphasizing time-to-value through rapid iterations, implementing CI/CD to support frequent safe releases, and using measurable evaluation criteria (especially time-to-charge and booking conversion). It remains focused on engineering workflow, real-time data integration, and scalability readiness rather than research-driven novelty.